Employee Workforce Analytics using Excel

Project Overview

This project focuses on analyzing employee data to understand workforce demographics, salary distribution, department-wise performance, and organizational trends.

The analysis helps HR teams make data-driven decisions related to hiring, performance evaluation, and workforce planning.

Project Objectives

- Analyze workforce demographics
- Understand employee performance patterns
- Support HR decision-making
- Enable predictive workforce insights
- Maintain structured employee records

Dataset Description

The dataset contains employee-related information within an organization, including personal, job, and performance details.

It is used for:

- HR analytics
- Workforce planning
- Performance evaluation
- Organizational development

Column Description

Column Name	Description
Employee ID	Unique identifier for each employee
Name	Full name of employee
Department	Department (HR, Finance, IT, Marketing, Sales)
Country	Employee location

Column Name	Description
Salary	Employee salary
Date of Joining	Joining date for tenure analysis
Performance Rating	Performance score (1–5 scale)
Gender	Employee gender
Email	Official email ID

Data Preparation (Excel)

- Removed duplicate records
- Handled missing values
- Standardized department names
- Formatted date column (Date of Joining)
- Ensured consistent salary format
- Validated performance rating scale

Analysis Performed

- **Descriptive Analysis**
  - Sales department has the highest salary levels
  - Most employees are from India
  - Marketing department shows highest performance ratings
  - Employee hiring has increased year over year
  - India has the highest salary distribution
- **Diagnostic Analysis**
  - Marketing employees show higher performance due to revenue contribution
  - Finance department has lower salaries and lower performance ratings
  - Salary levels influence performance trends across departments
- **Predictive Analysis**

- Marketing performance may decline if not optimized
- Salary structure impacts employee performance behavior

🔹 **Prescriptive Analysis**

- Focus on strengthening sales department to improve growth
- Increase salary in marketing to improve performance
- Improve finance department engagement and incentives

Key Insights

- 📌 Sales department has highest salary distribution
- 📌 Marketing employees have highest performance ratings
- 📌 Majority employees are from India
- 📌 Employee hiring trend is increasing yearly
- 📌 Salary directly influences performance in some departments

Business Recommendations

- Improve salary structure in Finance department
- Invest more in Sales and Marketing teams
- Provide performance-based incentives
- Focus on employee retention strategies
- Strengthen HR analytics for workforce planning

Conclusion

This Excel-based employee analysis provides valuable insights into workforce structure, salary distribution, and performance trends.

The dataset supports HR decision-making by identifying high-performing departments, salary gaps, and workforce growth patterns. It helps organizations improve productivity and optimize employee management strategies.